Trainer key | Pilot RCT and basic economic evaluation

Trainer copy. Do not distribute before participants complete the worksheet.

Core model numbers

The pilot RCT assigned **200 eligible businesses** to GreenWaste and **200** to control in one district. At 12 months, the treatment-minus-control difference in annual waste cost was **-1,014 AED** (95% interval **-1,167 to -862 AED**). The model treats the corresponding **1,014 AED annual saving** as the year-1 benefit and assumes it persists through years 2 to 5. The **1,800 AED** year-0 delivery cost and **5%** rate are fictional assumptions.

Present-value benefits **4,392 AED**; cost **1,800 AED**; benefit-cost ratio **2.44**; net present benefit **2,592 AED**. The ratio is a limited monetary comparison of an authority cost with a business financial saving. It is not a full social CBA or an authority budget return.

Q1 | Evidence and assumptions

a. The RCT estimates annual waste-management cost at **12 months** for eligible **pilot-district businesses**, relative to the lottery control group. Use -1,014 AED as a cost difference or +1,014 AED as a saving. (b) The 1,800 AED delivery cost is an **assumption**. Request programme accounts, scope, quantities and timing. (c) Years **2 to 5** have no RCT follow-up evidence in this case.

Q2 | Method and unit

a. CEA in AED per tonne requires a credible incremental **tonnes of landfill avoided** and complete incremental cost for the same alternative, perspective and period. The 1,014 AED waste-cost saving cannot become tonnes. (b) CUA typically uses **QALYs**; the case has no measured utility outcome. (c) CBA expresses benefits and costs in **money**. The worked ratio is partial because the perspective mixes actors and important entries remain unmeasured.

Q3 | Perspective

The assumed delivery cost falls on the **authority**; the measured waste-cost saving accrues to **businesses**. Accept a specific missing resource or effect such as staff time, business maintenance or environmental impact, together with a plausible record and timing. Check for transfers and double counting.

Q4 | Calculation

Year-1 present value: **1,014 / 1.05 = 966 AED**. Benefits **4,392 AED**; cost **1,800 AED**; ratio **4,392 / 1,800 ≈ 2.44**; net **4,392 - 1,800 ≈ 2,592 AED**. Multiplication by five ignores discounting, and the later four years are projections rather than observations.

Trainer key | Interpretation and advice

Q5 | Two thresholds

The model breaks even at roughly **416 AED** annual saving under its assumed cost, duration and rate. The pilot RCT interval on a saving scale is **862 to 1,167 AED**. It lies above the *fixed-model* break-even level but crosses the separate **1,000 AED policy minimum**. Higher costs, shorter persistence or a different discount rate can move the model threshold. The RCT interval is not an interval for the ratio.

Q6 | Scenarios and evidence

Among one-input changes, **30% annual fading** lowers the ratio most under the stated settings: **2.44 to 1.40**. Extra 600 AED cost gives **1.83**; an 8% rate gives **2.25**; three saving years give **1.53**. Fading plus 800 AED extra cost gives **0.97**, just below one. These are stress tests, not evidence about actual fading or costs. Useful requests include later cost outcomes, programme cost accounts, maintenance and staff-time records, and applicable local discount-rate guidance.

Q7 | Correct the claim

Accept wording such as: “The pilot RCT estimates about **1,014 AED lower annual waste cost at 12 months** for eligible businesses in one district. A partial model using an **assumed 1,800 AED delivery cost**, five modelled saving years and a 5% rate gives a ratio of **2.44**. We need cost records, persistence evidence and a reason to expect the pilot effect elsewhere before using it for citywide approval.”

The RCT does not prove the effect in every later year. The ratio does not establish full value for money or a citywide effect. The policy minimum remains uncertain because the RCT interval crosses 1,000 AED.

Q8 | Conditional advice

Accept different actions if the evidence and consequence are clear. A strong answer states that the limited model favours GreenWaste under its assumptions, then asks for a **specific fact** that could change the decision. For example: “I would prepare a phased extension while auditing per-business delivery costs and collecting another year of follow-up. If costs rise materially or savings fade, I would recalculate the ratio before wider rollout.” Do not require that exact recommendation.

The deck’s presenter notes contain the 90-minute facilitation sequence. The later mock report uses a separate city DiD model with a ratio of 1.86 and different timing assumptions. Do not compare that ratio with 2.44 as though it were the same calculation. This key is for checking worksheet responses, not a substitute for H7-A.